

## CHAPTER FIVE

# Market

*U.S. News and World Report* publishes an annual ranking of the states in the union. To sort the list, the magazine compiles scores on a range of factors, including quality of “health care, education, the economy, infrastructure, opportunity, fiscal stability, crime and corrections, and the natural environment.” In 2019, Washington took the top spot. In forty-eighth place was Mississippi, which scored near the very bottom in many categories, including health care, education, the economy, and infrastructure.<sup>1</sup> That Mississippi would score low on an arbitrary list of states is unsurprising—so much so that residents of neighboring states are quick to sigh “Thank God for Mississippi” when confronted with ranked lists of states, because their states frequently rank above Mississippi.<sup>2</sup>

But in perhaps a somewhat unexpected result, when the measure is homelessness, Mississippi is alone at the head of the class. That a state known for its poverty, poor educational system, and inferior health care would lead the nation with the lowest per capita of homelessness seems to strain credulity. But Mississippi

is also cheap. Evan Horowitz, in an article in the *Boston Globe*, analyzed the income and homelessness relationship and reached a similar conclusion: “Rich states have more homeless people. It’s not an iron-clad rule, just a loose tendency, but it cuts sharply against the grain of economic expectation.”<sup>3</sup> “Homelessness is largely about the price of homes,” he writes. “Mississippi and Alabama are cheap places to live.” With the invisible hand at work, states with lower median incomes tend to see cheaper housing costs—and vice versa. Washington, which enjoyed the top position in the *U.S. News and World Report* ranking, had at the time the fifth-highest per capita rate of homelessness in the country. Washington and Mississippi are far from outliers. Instead, they help to begin to illustrate the fundamental argument of this book: Homelessness is a housing problem.

Earlier chapters suggest that neither individual factors nor local culture and politics explain regional variation in rates of homelessness. We now turn to more fundamental structural drivers of this phenomenon: housing market conditions. In contrast to popular narratives concerning the causes of homelessness, abundant academic research provides credible links between housing markets and housing instability. As a result, a challenge has arisen to reconcile seemingly contradictory evidence—between anecdote and research. The complexity (and general inaccessibility) of the academic evidence base doesn’t help. Broadly speaking, scholarly communication has failed to inform or shift public perception of the nature of the country’s homelessness crisis, thereby opening the door for more readily interpretable (and emotional) explanations like drug use and mental health. But we argue the contradictions in question only emerge if we blur units of analysis. Throughout this book, we’ve sought to stress that attributing city-level findings to

individuals—or vice versa—will often lead to imprecise (or flat-out wrong) diagnoses. At the level of the city (or county), we’ve demonstrated that individual and cultural explanations fail to explain regional variance, even where they might offer explanations for why a given person lost their housing. Risk factors don’t imply root causes.

We now turn to a range of housing market explanations.

#### HOUSING COST BURDEN

A common narrative deployed to explain homelessness draws on the concept of housing cost burden: the percentage of one’s household income that goes toward rent. The higher your burden, the greater your risk of homelessness. This logic generally goes uncontested—and for good reason. For example, in the 2016 *State of the Nation’s Housing*, published by the Joint Center for Housing Studies at Harvard University, the authors note that “housing cost burdens are a fact of life for a growing number of renters. These burdens put households at risk of housing instability and homelessness, particularly in the nation’s high-cost cities.”<sup>4</sup> Other researchers have identified specific thresholds for median housing cost burdens that, upon crossing, boost rates of homelessness in cities.<sup>5</sup> And these findings make sense. There is little doubt that higher rent burdens increase precarity for vulnerable households.

It’s worth investigating, though, the extent to which housing cost burdens can explain the intercity variation in rates of homelessness currently under the microscope. To test these relationships, we first consider the association between median gross rent as a percentage of household income (among renters) and per capita rates of homelessness. As Figure 18 illustrates, there

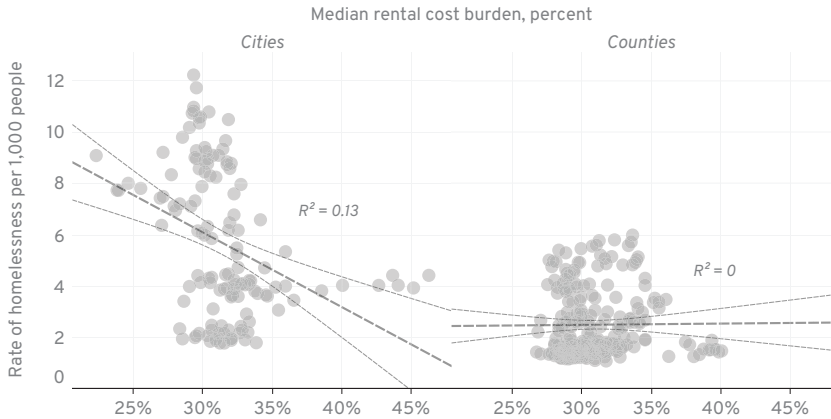


Figure 18. Median gross rent as a percentage of household income versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto median rental cost burden between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

is no clear correlation between these variables. In fact, in our data set, the relationship is mildly negative. This finding seems counterintuitive.

Part of what's going on here is that there is a tight relationship between housing costs and incomes within a given metropolitan area. Consider the distribution of median rent burdens: The vast majority of cities and counties in our sample have median rental burdens between 26 and 34 percent. The tight distribution demonstrates that rents tend to scale with incomes, and, therefore, even places with high median housing costs do not *necessarily* have high cost burdens. Cost burdens in San Francisco and Seattle are near the national average, and some of the highest median cost burdens for renters are found in lower-cost cities like Detroit. The story in Detroit and other postindustrial cities is one of low rents coupled with very low incomes—so low

that we observe relatively high housing cost burdens for renters. When combined with the low rates of homelessness, a city like Detroit helps to explain the slightly negative (but otherwise unsubstantial) relationship between cost burdens and homelessness. And, as always, note that we're interested in inter-regional variation here. For a given household, a high cost burden indisputably increases the risk of homelessness.

A critic might argue that *median* rent burden offers an inappropriate measure when studying homelessness, not least because households at the median are at very little risk of homelessness. A more appropriate question might be whether implied housing cost burdens faced by low-income households predict homelessness.<sup>6</sup> Figure 19 depicts the relationship.

Again, this relationship fails to provide a credible explanation. Low-income renters in Philadelphia have higher cost burdens (54 percent across the whole sample) than similarly situated low-income renters in New York (50 percent), San Francisco (40 percent), and Seattle/King County (40 percent). By every definition of the word, these burdens are unaffordable. And some renters have it worse, putting up to 70 or 80 percent of their incomes to housing costs. But if the housing market is to blame for regional variation in rates of homelessness, this common housing fraction—frequently blamed for homelessness—doesn't provide the explanation we seek.

Recent research offers more nuance. Chris Glynn and colleagues used data from all CoCs in the country—a much broader sample than the one we use in this book—and found evidence of inflection points in the relationship between housing cost burdens and community levels of homelessness.<sup>7</sup> According to their analysis, rates of homelessness rise sharply once housing cost burden exceeds 30–34 percent. This threshold makes

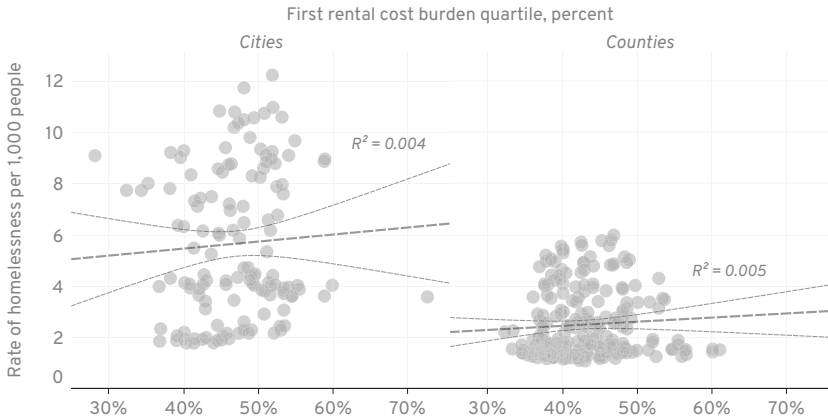


Figure 19. First-quartile housing cost burden versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto the first rental cost burden quartile between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

sense, especially given the 30 percent cutoff often deployed to signify whether a household is indeed “cost burdened.” Importantly, the researchers’ analysis employed statistical methods to control for variation along dimensions other than housing cost burden, including population size and poverty rates, in addition to allowing for the relationship between these variables and the outcome variable of interest—rates of homelessness—to be nonlinear. Doing so allowed the researchers to isolate the effect of a given variable (in this case, housing cost burden) while holding others constant and to identify the existence of the inflection points in question. Certainly, we agree that higher cost burdens lead to higher rates of homelessness at the individual level. And if we control for other factors, within a given community, rising housing cost burdens (especially past the inflection points identified by Glynn et al.) will lead to higher community-level

homelessness. Within our sample of thirty CoCs (representing most of the largest MSAs in the country), though, housing cost burden doesn't help explain regional variation.<sup>8</sup>

#### ABSOLUTE RENT LEVELS

As demonstrated above, the housing cost burden fraction (rent divided by income) fails to explain regional variation in rates of homelessness. But let's deconstruct the fraction into its component parts. In chapter 3, we tested a meaningful correlate of the denominator and found a negative relationship between poverty and rates of homelessness. High poverty regions do not have high rates of homelessness. Poverty may lead to homelessness at the individual level, but it does not explain regional variation. We now consider the numerator of the burden fraction: housing costs. Do *absolute* rent levels explain variation?<sup>9</sup>

Unsurprisingly, the landscape of our sample illustrates substantial variation in rents across CoCs. Consistent with expectations, a number of cities in the industrial Midwest—Detroit, Indianapolis, St. Louis, Cuyahoga County (Cleveland), and Hamilton County (Cincinnati)—have low median rents (below \$750). On the other end of the spectrum are a handful of expensive coastal cities and counties. Boston, San Francisco, King County (Seattle), San Diego County, and Washington, D.C., all see median monthly rents between \$1,500 and \$2,000. The highest rental costs in our sample are in Santa Clara County (Silicon Valley), where median rents are nearly \$2,300/month; over three times the level of Rust Belt cities. These stark differences highlight the radically different consequences of poverty depending on where one lives.

Consider the graphs in Figure 20. Unlike many previous analyses presented in this book, the relationship between the

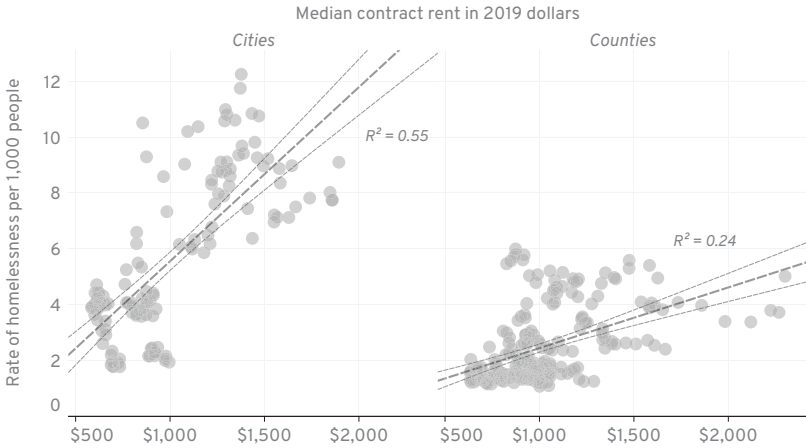


Figure 20. Median contract rent versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto median contract rent between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

variables in question is convincingly positive—and captures the intuitive relationship between housing markets and homelessness that we might expect at first blush. In city CoCs, the relationship between rents and homelessness is drastic; in county CoCs (i.e., regions with less urban concentration), the relationship is substantial but slightly less pronounced. Our measure of explained variation ( $R^2$ ) is significantly higher than we’ve observed elsewhere.

In Figure 21, we also consider the relationship between rents at the twenty-fifth percentile and rates of homelessness. Precariously housed adults and families rarely rent at the fiftieth percentile, so just as was the case in our test of housing cost burden, it’s worth examining low-income rents. As the graphic demonstrates, the story is the same. Places with higher rents at the twenty-fifth percentile have higher rates of homelessness.<sup>9</sup>

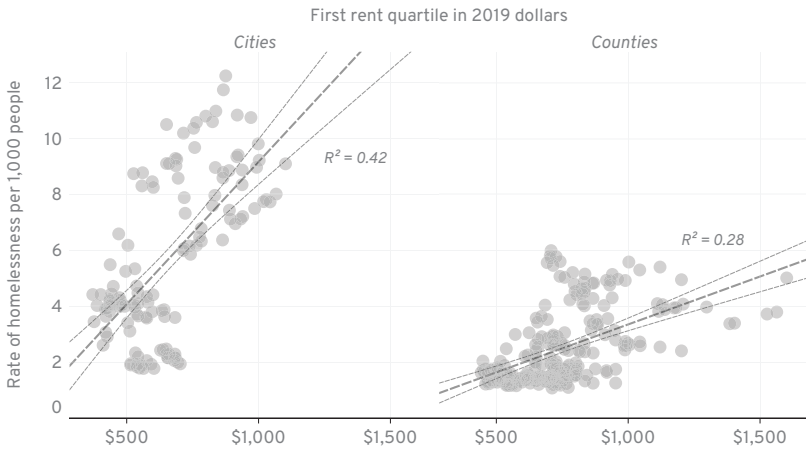


Figure 21. First-quartile rent versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto the first rent quartile between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

These findings invite an obvious question: Why do absolute rents explain regional variation, but housing cost burden doesn't? One plausible explanation comes from the fact that for households with extremely low incomes, the absolute level of rent is far more important. If you're a cost-burdened renter at the upper end of the income spectrum, you can choose to spend less on housing if the need arises. If you're a cost-burdened renter at the lower end of the income spectrum, that's not true: Rents don't drop to zero. Instead, they hit a floor determined by the local housing context and the rules that govern housing quality; eventually, you're living in the cheapest housing available (and we're left asking a question about low-income housing supply).

This part of the story is a double-edged sword. On a relative basis, the quality of housing in the United States is very high. Housing units of inadequate quality are rare, and levels

of quality don't differ meaningfully between subsidized and market-rate units.<sup>10</sup> This national move toward high-quality housing unquestionably represents a positive change relative to prior eras, when housing quality represented a major public concern.<sup>11</sup> At the same time, because of the well-documented link between housing quality and price, increased quality—in a somewhat perverse result—may limit the supply of “naturally occurring” affordable rental housing due to higher prices and may increase the cost of new, affordable units.<sup>12</sup> Certainly, rules must govern housing quality and safety. But this contradiction highlights a logical inconsistency in the U.S. approach to affordable development: We mandate a minimum level of quality but fail to provide sufficient resources to households who can't afford the housing constructed or maintained at this standard.

In sum, we believe that this finding—the importance of absolute rents—speaks to a basic and intuitive fact: For a highly impoverished household, it is likely easier to access housing in Detroit or St. Louis, where median rents are between \$600 and \$700 per month, than in San Francisco and Santa Clara County, where costs are three to four times higher. Affordable housing that meets basic standards is accessible in some locations, but not others. There are more ways to “make it work” in St. Louis. Minimum wage labor, public assistance, and support from family and friends can be enough to help you get by. Such supports—which also exist in more expensive cities—are woefully inadequate given the cost of housing.

#### RENTAL MARKET VACANCY RATES

The final variable we investigate is rental market vacancy rates—that is, the proportion of available rental units in a given region

at any given point in time. In 2019, numerous CoCs had rental market vacancy rates in excess of 8 percent, including Atlanta, St. Louis, Mecklenburg County (Charlotte), Dallas County, Clark County (Las Vegas), and Hillsborough County (Tampa)—representing a more accommodating rental market for housing. That same year, the availability of rental housing was severely constrained in several markets, among them the list of usual suspects, including San Francisco, Boston, New York, Santa Clara County, Los Angeles County, and King County (Seattle). But there's a surprising addition to this list, as well: Detroit.

In the decade since the Great Recession, rental vacancies in Detroit have fallen considerably. In 2010, Detroit had a rental market vacancy rate of nearly 14 percent: the highest rate in our sample. Over the ensuing nine years, the vacancy rate fell by over ten percentage points to less than 4 percent, placing Detroit in the same company as San Francisco and New York. Explaining this precipitous decline requires a closer look at post-recession Detroit. Indeed, both total population and the total number of housing units have fallen in the city since 2007. Most substantial, though, are the city's levels of unoccupied, vacant housing units. Over this period, between 25 and 30 percent of all units were unoccupied. To understand the city's tight rental market in the context of a quarter of all housing units being vacant, it is important to note that most of these unoccupied units are not available for lease. Many units are abandoned—and therefore not included in the active housing stock in the city. As population and employment levels have stabilized in Detroit over the last decade, housing has become scarcer; and serious concerns have arisen regarding rising housing instability and homelessness in the years to come.<sup>13</sup> Through 2019, rental costs remained low, but if rental housing scarcity continues to persist, we'd

anticipate upward pressure on rents—which will further punish many low-income households residing in this city. Detroit has a hidden advantage, however, that few other cities with tight rental markets enjoy: Detroit’s vacant units offer the potential for refurbishment and could then be brought back into the housing system. As Robin Runyan wrote in *Hour Detroit*, “With nearly 81,000 off-market vacant units, the city of Detroit does have sufficient housing stock.”<sup>14</sup>

Outside the somewhat anomalous case of Detroit, we observe a strong, somewhat intuitive relationship between vacancy rates and rental costs (see Figure 22). Because rents tend to be higher when vacancy rates are low (all else being equal), we’re ultimately measuring different aspects of a related phenomenon; these variables aren’t independent of one another. This relationship is well known and has been observed by researchers in the fields of economics and real estate for decades.<sup>15</sup> As it relates to the topic of homelessness, given the relationship between two variables, we would expect to see higher rates of homelessness where vacancy rates are low.

Indeed, graphically and statistically, as shown in Figure 23, vacancy rates join rental costs as the only variables that explain regional variation in homelessness.<sup>16</sup> Because these variables are intimately related—vacancy rates help to predict housing costs—we don’t attempt to rank the variables in terms of their relative impact on rates of homelessness.<sup>17</sup> Rather, we present these analyses as two examples of how related housing market dynamics explain why some regions have high rates of homelessness while others don’t. While beyond the scope of the base analyses in this book, we also conducted a series of multivariate models—statistical analyses that attempt to explain variance in per capita rates of homelessness using more than one



Figure 22. Median rent versus rental vacancy rate. Dashed lines indicate a linear regression of rental vacancy rates onto media contract rent between 2007 and 2019 for a sample of U.S. regions. Data source: U.S. Census Bureau

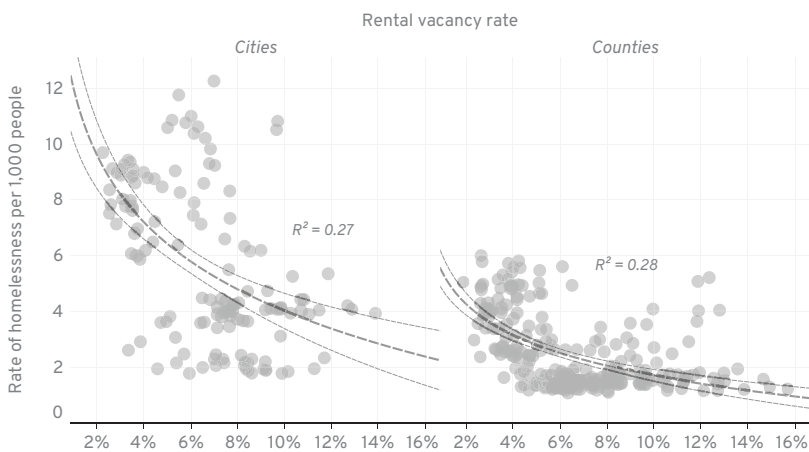


Figure 23. Rental vacancy rate versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto the natural log of rental vacancy rate between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

explanatory variable at once—that demonstrate similar results. Rents and vacancies continue to have significant explanatory power even after controlling for the effects of a range of different variables, including poverty, income, housing cost burden, population, age, race/ethnicity, gender, household structure, housing tenure, and inequality. Generally speaking, these housing market effects are strongest when seeking to explain indexed rates of per capita homelessness (i.e., when we place counties and cities on equal footing). They also tend to persist in so-called area fixed effects models whereby we seek to explicitly understand intra-regional variation over time. Comparatively speaking, these multivariate and fixed effects models suggest that vacancy rates appear to persist more strongly intra-regionally and rent effects persist more strongly inter-regionally.

To further explore the relationship, the following graphic plots first-quartile rental market vacancy rates against per capita rates of homelessness (Figure 24). (First-quartile vacancy rates come from measuring the vacancy rate of the market for units with rents between zero and the twenty-fifth percentile of regional rent.) Unsurprisingly, vacancy rates of cheaper rental units predict rates of homelessness, as well. While vacancies for these units appear to have slightly less predictive power than the overall market vacancy rates, because the difference is modest, one shouldn't draw too many conclusions here. The point is that the housing market—as a whole—helps create the conditions in which homelessness varies from region to region. It's not merely a shortage of low-income housing: It's an overall housing shortage that matters. As we discuss in the final chapter of this book, that fact doesn't imply that exclusively building luxury condos will solve a region's housing challenges—far from it. Housing doesn't magically “filter” or trickle down to low-income households. It is essential, therefore,

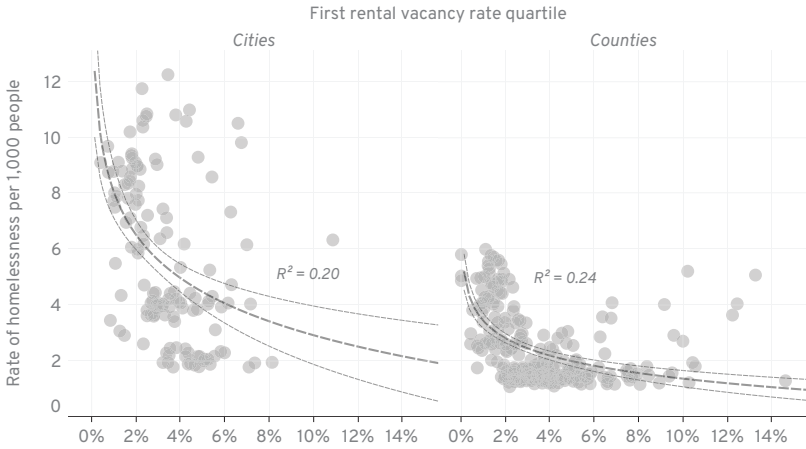


Figure 24. First-quartile vacancy rate versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto the natural log of the first rental vacancy rate quartile between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

to ensure sufficient housing for the lowest-income households. But, as these figures demonstrate, tight housing conditions persist throughout the income spectrum and therefore, boosting the production of housing at all levels will relieve some of the pressure in these tight, expensive housing markets.

#### UNDERSTANDING THE MARKET

If we attribute regional variation in homelessness to housing market dynamics—rent levels and vacancy rates—as the previous analysis suggests we ought to, we need to understand why some cities have high housing costs and low vacancy rates while others do not. To do so, we need to understand how the forces of supply and demand shape the housing stock in a particular city

or region. In the next chapter, we provide a more detailed analysis, but we provide a brief discussion here.

It is common among people who subscribe to structural explanations of homelessness to draw a link between population growth and homelessness. This connection makes intuitive sense. Rapid increases in employment and population may bring wealthy people to a community and displace existing residents. This problem is one of demand: As population and wages increase, the demand for housing grows. Indeed, housing scholars identify the key drivers of housing demand as income, population, and household formation.<sup>18</sup> Given the employment and population booms in cities like San Francisco, Seattle, and Washington, D.C., it would be easy to blame increasing rates of homelessness on these demand dynamics. But further investigation suggests that focusing solely on the demand side of the equation is inadequate. Many cities and counties with high rates of population growth fail to see high per capita rates of homelessness. In our sample, we observe—in Figure 25—only a modest positive relationship between population growth and homelessness in cities and virtually no relationship in counties.<sup>19</sup>

While the correlations in the previous chart tell a broad story about the population-homelessness relationship (or lack thereof), the case of Mecklenburg County (Charlotte), North Carolina, offers an illustration of a rapidly growing metropolitan area where rising demand has not produced scarce housing (when compared to other high-growth cities). Driven by a diverse economy, the population of Mecklenburg County grew by 28 percent from 2007 to 2019—one of the fastest growing large counties in the country. By comparison, King County (Seattle), home to a tech-driven population boom, grew by only 21.5 percent over the same period. The growing population of

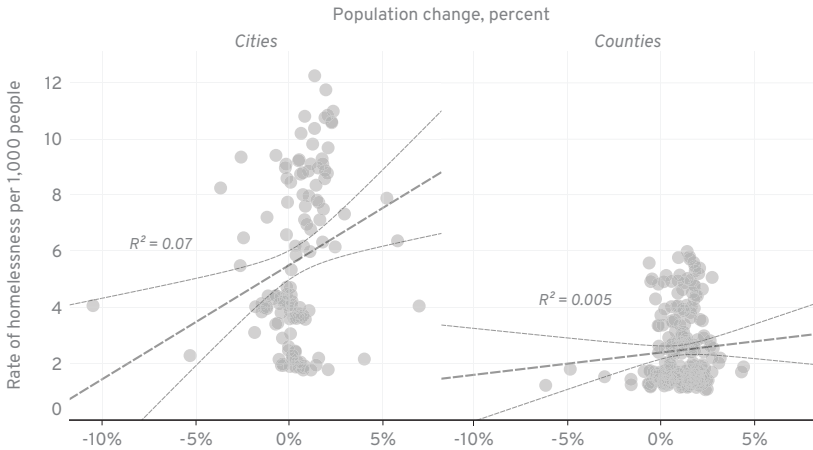


Figure 25. Change in population versus PIT count (per capita). Dashed lines indicate a linear regression of per capita PIT counts onto population change between 2007 and 2019 for a sample of U.S. regions. Data source: HUD / U.S. Census Bureau

the Charlotte region works for large employers in financial services, health care, and retail. And indeed, population growth has created a sharp increase in the demand for local housing. But while rents have risen during this expansion, vacancy rates have remained elevated, thanks to the quick and substantial construction of new housing. Over the period covered in our study, the average vacancy rate in Mecklenburg County hovered near 7 percent, and the number of occupied rental housing units increased by over 43 percent—a dramatic increase. Given sufficient housing resources, rates of homelessness have remained relatively low compared to many other fast-growing regions in the country. Over the study period, rates of homelessness in Mecklenburg have ranged from a low of 1.4 per thousand (in 2017) to 3.1 (in 2010)—well below the rates of other

fast-growing counties, such as Multnomah (Portland) and King (Seattle).

Why, in response to substantial population growth, was Charlotte able to build a substantial number of new housing units? And why do other boomtowns like San Francisco and Seattle face chronic shortages of housing, even as population and employment continue to rise? More generally, why do some cities respond to growth by adding additional housing units, while other cities continue to operate with limited supplies of housing? To understand what's going on here, it's critical to recognize the concept of housing supply elasticity. In economics, price elasticity of supply refers to how responsive ("elastic") the supply of a given good or service is to a change in its price. The price elasticity of supply of housing, then, measures the change in housing supply in response to a change in the price of housing. When prices increase significantly, an elastic market will, in response, construct a large amount of housing, while an inelastic market will produce relatively fewer units. Urban economists have calculated local housing supply elasticities for cities across the United States, and cities with inelastic housing supply tend to have significant topographical constraints (like water and mountains) and/or restrictive regulatory environments that prevent rapid construction.<sup>20</sup> Large cities with inelastic housing supplies include New York City, Los Angeles, San Francisco, Seattle, and Miami. The urban economist Albert Saiz lists Houston, Austin, Charlotte, Kansas City, and Indianapolis as cities with the most elastic housing supply in the country.

In inelastic markets, increases in the demand for (and prices of) housing won't result in a robust supply response, thereby exacerbating existing housing market pressures. The relatively elastic housing supply of Charlotte, on the other hand—driven

by an accommodating topography and a less restrictive regulatory environment—helps explain why, in response to a local economic boom, construction increased substantially, housing costs failed to spike to the extent they have in other regions, and rental market vacancies remain higher than in many other cities experiencing booms.

Because of the importance of rental market vacancy rates in this analysis, it is important to investigate the concept more deeply. Even in places with low vacancy rates, many units are often vacant. (We saw one version of this phenomenon in Detroit above.) A recent article out of the Bay Area proclaimed: “San Francisco has nearly five empty homes per homeless resident.”<sup>21</sup> This fact feels like a contradiction—and at face value, it is. And while the San Francisco calculation is technically correct—the numbers are right there on the American Community Survey (ACS)—it points to a central challenge of operationalizing the concept of vacancy: that of how to measure the idea in a manner that captures the housing market to which a given renter might actually be exposed. To shed light on this question, let’s break down vacant units in San Francisco using data from the ACS.

The “vacant” units referenced by the author of the San Francisco article aren’t vacant in the sense that a potential buyer or renter could secure one of them. Many are already rented or purchased but remain vacant for a portion of the year; others are seasonal or vacation homes. The ACS offers a catch-all “Other Vacant” category, too, which includes units “held for occupancy by a caretaker or janitor, and units held for personal reasons of the owner.”<sup>22</sup> There’s undoubtedly a story of excess and waste to be told here, but it is a story of vacancy writ large—and not one of rental vacancies. Table 1 shows the actual breakdown of vacant units in San Francisco in 2017.

TABLE 1  
Summary of Vacant Units: San Francisco, 2017

|                                                  |         |
|--------------------------------------------------|---------|
| Total Housing Units                              | 397,566 |
| Occupied Units                                   | 360,323 |
| <i>Owner-occupied</i>                            | 131,576 |
| <i>Renter-occupied</i>                           | 228,747 |
| Vacant Units                                     | 37,243  |
| <i>For rent</i>                                  | 8,292   |
| <i>Rented, not occupied</i>                      | 1,428   |
| <i>For sale, only</i>                            | 1,244   |
| <i>Sold, not occupied</i>                        | 4,157   |
| <i>Seasonal, recreational, or occasional use</i> | 10,603  |
| <i>Other vacant</i>                              | 11,519  |
| Homeowner Vacancy Rate                           | 0.9%    |
| Rental Vacancy Rate                              | 3.5%    |

As the table demonstrates, for potential renters in San Francisco, there are a limited number of options. The rental vacancy rate is calculated by dividing the total number of rental units available for rent (8,292) by the sum of that value and the number of occupied rental units (228,747). The quotient produces the rental market vacancy rate—in this case, 3.5 percent.

In San Francisco, more than twenty-two thousand units are vacant for “Seasonal” or “Other” reasons. Thus, by the broad definition of “vacant,” there are enough units to house all people experiencing homelessness in the city, but that’s not the same thing as saying San Francisco has sufficient housing for all of its residents and it just needs to allocate the housing more efficiently. In reality, the number of *rentable* rental units is remarkably low. We can’t efficiently allocate our way out of this problem.

Still, 8,292 potential units available for rent sounds like a large number. Why so many rental vacancies in a city with

such significant demand for housing? The answer lies in studies of housing markets. Scholars have repeatedly identified a “natural vacancy rate” for rental housing between 4 and 5 percent.<sup>23</sup> Just as economists consider there to be a natural rate of unemployment—below which there is a shortage of labor—the natural vacancy rate indicates the level at which the rental market is in equilibrium. Given the normal frictions and transitions of any housing market, it is impossible for every unit to be occupied at once. People move and units are vacated, occasionally for an extended period. Vacancy rates above the natural rate indicate a market with surplus units, while markets operating below the natural rate have a housing deficit. Rather than suggesting an abundance of housing units ready to be occupied by the precariously housed, San Francisco’s 3.5 percent rental market vacancy rate indicates a market in deficit—in dire need of additional rental units to meet the needs of its growing population.

Urban economics literature helps provide an important link between the two variables of interest in this chapter: rental market vacancy rates and absolute levels of rents. Dozens of studies have highlighted a strong theoretical and empirical relationship between these two variables. According to Kenneth Rosen and Lawrence Smith, variations “in the vacancy rate around some natural rate of vacancy exert a significant influence on the rate of change of the price of rental housing services.”<sup>24</sup> Subsequent research has refined Rosen and Smith’s analysis but reached consistent conclusions.<sup>25</sup> The story is straightforward enough. When housing market conditions are tight and vacancies are limited, rents in a region receive upward pressure. Accordingly, the absolute rent levels tested in this chapter can be partially explained by the prevailing vacancy rates in a given locale (and not necessarily the other way around).

The housing market factors outlined in this chapter—in particular, absolute rents and vacancy rates—provide a credible explanation for regional variations in homelessness. As noted previously, this finding does not suggest that individual factors do not matter. Instead, it suggests that household risk factors may produce a spell of homelessness in some settings and circumstances but not in others. Context matters. Vulnerable households live in every city of the country; the difference in *rates* of homelessness can be attributed to structural factors associated with the housing market. Individual factors may help explain who becomes homeless in a tight housing market like Seattle, but they fail to explain why the city has five times the rate of homelessness of Chicago.

It's worth noting that while these market factors help to explain the differences we're after, they don't capture everything. Statistically, even after taking housing into account, plenty of meaningful regional variation remains unexplained. But it would be shocking if rents explained 100 percent of variance. Homelessness is a complex social phenomenon involving the lived experiences of hundreds of thousands of people; it is not a function of one or two variables. (If it were, policy solutions to housing instability would likely be more readily available.) In other words, by no stretch of the imagination do housing market variables explain all variation in the manifestation or experience of homelessness across cities around the country. Certainly, other research examining within-area changes over time has identified a positive relationship between rents and homelessness.<sup>26</sup> And yet even within a given city, there may be circumstances in which housing costs rise and the rate of homelessness does not. Brendan O'Flaherty highlights two such periods—one in New York, and one in Los Angeles—where such an outcome occurred.<sup>27</sup>

But these market factors are still useful. That so much regional variance is captured by a handful of straightforward measurements suggests there's something fundamental at play here. And to the extent that cities and counties are in a position to affect structural change, understanding these fundamental relationships remains critical to crafting the most appropriate response to the crisis.

Our analysis can also consider other upstream explanations for the variation that we observe. One such structural factor that has been implicated as a potential driver of homelessness is income inequality. Existing research demonstrates that inequality helps to explain both homelessness and the vastly different housing market conditions that we observe throughout the United States. As described in our first chapter, O'Flaherty highlighted the relationship between income inequality and homelessness in his 1996 book. Importantly, he drew a link between income inequality and the housing market dynamics that produce high rates of homelessness.<sup>28</sup> While we argue in this book that housing market factors explain variation in rates of homelessness, income inequality helps us understand why housing market conditions are more conducive to homelessness in some locales than others. Recent research from Thomas Byrne, Benjamin Harwood, and Anthony Orlando provides just such a link. In a sample of 239 communities in the United States (a far larger sample than the one we include in our book), income inequality—as measured by Gini coefficients—has a *causal* relationship with rates of homelessness.<sup>29</sup> In other words, the pronounced income inequality seen in many U.S. cities helps to produce an environment with high housing costs and modest incomes where homelessness flourishes. In our sample of thirty large metropolitan areas, we do not find a strong link between

income inequality and per capita rates of homelessness—it is present in city CoCs, but not in county CoCs. This is likely due to the fact that many of the cities in our sample have relatively high rates of income inequality. In other words, there isn't enough variation here. The much larger sample used by Byrne and colleagues includes far greater variation in levels of inequality and therefore provides more opportunity to explore this relationship. We conclude that income inequality may not explain regional variation among the largest and most unequal cities (those metropolitan areas in our sample), but it does help to explain why homelessness is much worse in communities where income inequality is high.